

1. Let $y = y(x)$ be a solution of the differential equation $(x \cos x)dy + (xy \sin x + y \cos x - 1)dx = 0, 0 < x < \frac{\pi}{2}$. If $\frac{\pi}{3}y\left(\frac{\pi}{3}\right) = \sqrt{3}$, then $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right) + 2y'\left(\frac{\pi}{6}\right)\right|$ is equal to _____.
[2023 (06 Apr Shift 1)]
2. If the solution curve $f(x, y) = 0$ of the differential equation $(1 + \log_e x) \frac{dx}{dy} - x \log_e x = e^y, x > 0$, passes through the points $(1, 0)$ and $(a, 2)$, then a^a is equal to
[2023 (06 Apr Shift 2)]
(1) e^{2e^2}
(2) e^{e^2}
(3) $e^{\sqrt{2}e^2}$
(4) $e^{2e\sqrt{2}}$
3. If the solution curve of the differential equation $(y - 2 \log_e x)dx + (x \log_e x^2)dy = 0, x > 1$ passes through the points $(e, \frac{4}{3})$ and (e^4, α) , then α is equal to _____.
[2023 (08 Apr Shift 1)]
4. Let the solution curve $x = x(y), 0 < y < \frac{\pi}{2}$, of the differential equation $(\log_e(\cos y))^2 \cos y dx - (1 + 3x \log_e(\cos y)) \sin y dy = 0$ satisfy $x\left(\frac{\pi}{3}\right) = \frac{1}{2 \log_e 2}$. If $x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_e n}$, where m and n are coprime, then mn is equal to
[2023 (08 Apr Shift 2)]
5. Let f be a differentiable function such that $x^2 f(x) - x = 4 \int_0^x t f(t) dt, f(1) = \frac{2}{3}$. Then $18 f(3)$ is equal to
[2023 (10 Apr Shift 1)]
(1) 210
(2) 160
(3) 150
(4) 180
6. The slope of tangent at any point (x, y) on a curve $y = y(x)$ is $\frac{x^2 + y^2}{2xy}, x > 0$. If $y(2) = 0$, then a value of $y(8)$ is
[2023 (10 Apr Shift 1)]
(1) $-4\sqrt{2}$
(2) $2\sqrt{3}$
(3) $-2\sqrt{3}$
(4) $4\sqrt{3}$
7. Let the tangent at any point P on a curve passing through the points $(1, 1)$ and $(\frac{1}{10}, 100)$, intersect positive x-axis and y-axis at the points A and B respectively. If $PA : PB = 1 : k$ and $y = y(x)$ is the solution of the differential equation $e^{\frac{dy}{dx}} = kx + \frac{k}{2}, y(0) = k$, then $4y(1) - 5 \log_e 3$ is equal to _____.
[2023 (10 Apr Shift 2)]
8. Let $y = y(x)$ be a solution curve of the differential equation, $(1 - x^2 y^2)dx = ydx + xdy$, If the line $x = 1$ intersects the curve $y = y(x)$ at $y = 2$ and the line $x = 2$ intersects the curve $y = y(x)$ at $y = \alpha$, then a value of α is
[2023 (11 Apr Shift 1)]
(1) $\frac{1-3e^2}{2(3e^2+1)}$
(2) $\frac{1+3e^2}{2(3e^2-1)}$
(3) $\frac{3e^2}{2(3e^2-1)}$
(4) $\frac{3e^2}{2(3e^2+1)}$
9. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{5}{x(x^5+1)}y = \frac{(x^5+1)^2}{x^7}, x > 0$. If $y(1) = 2$, then $y(2)$ is equal to
[2023 (11 Apr Shift 2)]
(1) $\frac{637}{128}$
(2) $\frac{679}{128}$
(3) $\frac{693}{128}$
(4) $\frac{697}{128}$

10. Let $y = y(x)$, $y > 0$, be a solution curve of the differential equation $(1 + x^2)dy = y(x - y)dx$. If $y(0) = 1$ and $y(2\sqrt{2}) = \beta$, then

[2023 (12 Apr Shift 1)]

- (1) $e^{3\beta-1} = e(3 + 2\sqrt{2})$
- (2) $e^{3\beta-1} = e(5 + \sqrt{2})$
- (3) $e^{\beta-1} = e^{-2}(3 + 2\sqrt{2})$
- (4) $e^{\beta-1} = e^{-2}(5 + \sqrt{2})$

11. Let $y = y_1(x)$ and $y = y_2(x)$ be the solution curves the differential equation $\frac{dy}{dx} = y + 7$ with initial conditions $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then the curves $y = y_1(x)$ and $y = y_2(x)$ intersect at

[2023 (13 Apr Shift 1)]

- (1) no point
- (2) two points
- (3) one point
- (4) infinite number of points

12. If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{4x}{(x^2-1)}y = \frac{x+2}{(x^2-1)^{\frac{5}{2}}}$, $x > 1$ such that $y(2) = \frac{2}{9}\log_e(2 + \sqrt{3})$ and

$y(\sqrt{2}) = \alpha \log_e(\sqrt{\alpha} + \beta) + \beta - \sqrt{\gamma}$, $\alpha, \beta, \gamma \in \mathbb{N}$, then $\alpha\beta\gamma$ is equal to

[2023 (13 Apr Shift 2)]

13. Let $x = x(y)$ be the solution of the differential equation $2(y+2)\log_e(y+2)dx + (x+4-2\log_e(y+2))dy = 0$, $y > -1$ with $x(e^4 - 2) = 1$. Then $x(e^9 - 2)$ is equal to

[2023 (15 Apr Shift 1)]

- (1) 3
- (2) $\frac{4}{9}$
- (3) $\frac{32}{9}$
- (4) $\frac{10}{3}$

ANSWER KEYS

1. (2) 2. (1) 3. (3) 4. (12) 5. (2) 6. (4) 7. (5) 8. (2)
9. (3) 10. (1) 11. (1) 12. (6) 13. (3)

1. (2)

Given,

$$(x \cos x)dy + (xy \sin x + y \cos x - 1)dx = 0$$

$$\Rightarrow (x \cos x) \frac{dy}{dx} + (yx \sin x + y \cos x) = 1$$

$$\Rightarrow \frac{dy}{dx} + y \left(\frac{x \sin x + \cos x}{x \cos x} \right) = \frac{1}{x \cos x}$$

Which is a linear differential equation of form $\frac{dy}{dx} + P(x)y = Q(x)$
Now we know that integrating factor will be $e^{\int P dx}$

$$\Rightarrow IF = e^{\int \left(\tan x + \frac{1}{x} \right) dx} = e^{\ln |\sec x| + \ln x} = x \cdot \sec x$$

Hence, solution of linear differential equation is given by,

$$\Rightarrow y \cdot x \sec x = \int \frac{x \sec x}{x \cos x} dx$$

$$\Rightarrow xy \sec x = \tan x + C$$

Now using $\frac{\pi}{3} y \left(\frac{\pi}{3} \right) = \sqrt{3}$ we get,

$$\Rightarrow \frac{\pi}{3} y \left(\frac{\pi}{3} \right) \sec \frac{\pi}{3} = \tan \frac{\pi}{3} + C$$

$$\Rightarrow C = \sqrt{3}$$

Hence, $xy \sec x = \tan x + \sqrt{3}$

$$\Rightarrow xy(x) = 2 \sin \left(x + \frac{\pi}{3} \right)$$

Now differentiating both side we get,

$$\Rightarrow xy'(x) + y(x) = 2 \cos \left(x + \frac{\pi}{3} \right)$$

Now again differentiating we get,

$$\Rightarrow xy''(x) + 2y'(x) = -2 \sin \left(x + \frac{\pi}{3} \right)$$

$$\text{Thus } \frac{\pi}{6} y'' \left(\frac{\pi}{6} \right) + 2y' \left(\frac{\pi}{6} \right) = -2$$

$$\text{Hence } \left| \frac{\pi}{6} y'' \left(\frac{\pi}{6} \right) + 2y' \left(\frac{\pi}{6} \right) \right| = 2$$

2. (1)

The given differential equation is

$$(1 + \ln(x)) \frac{dx}{dy} - x \ln(x) = e^y$$

Put $x \ln(x) = t$

$$\Rightarrow (1 + \ln(x)) dx = dt$$

$$\Rightarrow \frac{dt}{dy} - t = e^y \text{ which is a linear differential equation.}$$

$$I.F = e^{-\int dy}$$

$$= e^{-y}$$

The solution of the differential equation is

$$\Rightarrow t \times e^{-y} = \int e^y \times e^{-y} dy + c$$

$$\Rightarrow t \times e^{-y} = y + c$$

$$\Rightarrow x \ln(x) = y e^y + c e^y$$

$$\text{Put } x = 1, y = 0$$

$$\Rightarrow c = 0$$

$$\text{Put } x = a, y = 2$$

$$a \ln(a) = 2e^2$$

$$\therefore a^a = e^{2e^2}$$

Hence this is the correct option.

3. (3)

Given,

$$(y - 2 \log_e x) dx + (x \log_e x^2) dy = 0, x > 1$$

$$\Rightarrow 2x \ln x \frac{dy}{dx} + y = 2 \ln x \text{ \{where } \log_e x = \ln x \}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{2x \ln x} = \frac{1}{x}$$

Which is linear differential equation,

$$\text{So, } I.F = e^{\int \frac{1}{2x \ln x}} = \sqrt{\ln x}$$

Now, solution of the equation is given by,

$$y \cdot \sqrt{\ln x} = \int \frac{\sqrt{\ln x}}{x} dx$$

$$\Rightarrow y \cdot \sqrt{\ln x} = \frac{2}{3} (\ln x)^{\frac{3}{2}} + C \quad \dots (i)$$

Given, eq. (i) passes through point $(e, \frac{4}{3})$

$$\text{So, } C = \frac{2}{3}$$

Hence, solution of differential equation will be,

$$y \sqrt{\ln x} = \frac{2}{3} (\ln x)^{\frac{3}{2}} + \frac{2}{3}$$

Also given, above equation passes through point (e^4, α)

$$\alpha \sqrt{\ln e^4} = \frac{2}{3} (\ln e^4)^{\frac{3}{2}} + \frac{2}{3}$$

$$\Rightarrow 2\alpha = \frac{2}{3} (4)^{\frac{3}{2}} + \frac{2}{3}$$

$$\Rightarrow \alpha = \frac{1}{3} \times 8 + \frac{1}{3}$$

$$\Rightarrow \alpha = 3$$

4. (12)

Given,

$$(\log_e (\cos y))^2 \cos y dx - (1 + 3x \log_e (\cos y)) \sin y dy = 0$$

$$\Rightarrow \frac{dx}{dy} + \left(\frac{-3 \tan y}{\ln \cos y} \right) x = \frac{\sin y}{(\ln \cos y)^2 \cdot \cos y}$$

Which is a linear differential equation,

$$\text{So, Integrating factor will be, } IF = e^{\int \frac{-3 \tan y}{\ln \cos y} dy} = (\ln \cos y)^3$$

So, solution of differential equation is given by,

$$x (\ln \cos y)^3 = \int \frac{\sin y}{\cos y} (\ln \cos y) dy$$

$$\Rightarrow x (\ln \cos y)^3 = \frac{-\frac{1}{2} (\ln \cos y)^2}{2} + c$$

$$\text{Now using the given value of } x\left(\frac{\pi}{3}\right) = \frac{1}{2 \ln 2}$$

We get, $c = 0$

$$\text{Hence, } x = \frac{-1}{2 \ln \cos y}$$

$$\text{Now finding, } x\left(\frac{\pi}{6}\right) = -\frac{1}{2 \ln \left(\frac{\sqrt{3}}{2}\right)} = \frac{1}{\ln 4 - \ln 3}$$

Now on comparing with $\frac{1}{\log_e m - \log_e n}$ we get, $m = 4, n = 3$

Hence, $mn = 12$

5. (2)

Given,

$$x^2 f(x) - x = 4 \int_0^x t f(t) dt$$

Now Differentiating both side w.r.t. x we get,

$$x^2 f'(x) + 2x f(x) - 1 = 4x f(x)$$

$$\Rightarrow x^2 f'(x) - 1 = 2x f(x)$$

$$\Rightarrow \frac{dy}{dx} - \frac{2}{x} y = \frac{1}{x^2} \quad \left(\text{Let } y = f(x) \frac{dy}{dx} = f'(x) \right)$$

Which is a linear differential equation,

$$\text{So, I. F} = e^{\int -\frac{2}{x} dx} = e^{-2 \ln x} = \frac{1}{x^2}$$

Now solution of the differential equation is given by,

$$\frac{y}{x^2} = \int \frac{1}{x^4} dx + C$$

$$\Rightarrow \frac{y}{x^2} = -\frac{1}{3x^3} + C$$

$$\text{Now given, } f(1) = \frac{2}{3}$$

$$\text{So, } \frac{2}{3} = -\frac{1}{3} + C$$

$$\Rightarrow C = 1$$

Hence, the function will be,

$$f(x) = -\frac{1}{3x} + x^2$$

So, the required value is given by,

$$18 f(3) = 18 \left[-\frac{1}{9} + 9 \right] = 160$$

6. (4)

Given:

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$$

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{1+v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1-v^2}{2v}$$

$$\Rightarrow \int \left(\frac{2v}{v^2-1} \right) dv = \int \frac{dx}{x}$$

$$\Rightarrow \log_e |v^2 - 1| = \log_e \left(\frac{C}{x} \right)$$

$$\Rightarrow \frac{y^2 - x^2}{x^2} = \frac{C}{x}$$

$$\Rightarrow y^2 - x^2 = Cx$$

Put $x = 2$ and $y = 0$ we get,

$$0 - 2^2 = 2C \Rightarrow C = -2$$

$$\Rightarrow y^2 = x^2 - 2x$$

$$\Rightarrow y(8) = \sqrt{8^2 - 16}$$

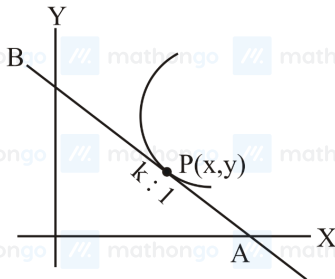
$$\Rightarrow y(8) = \sqrt{48} = 4\sqrt{3}$$

7. (5)

Given,

The tangent at any point P on a curve passing through the points $(1, 1)$ and $(\frac{1}{10}, 100)$, intersect positive x-axis and y-axis at the points A and B respectively,

And $P A : P B = 1 : k$ and $y = y(x)$ is the solution of the differential equation $e^{\frac{dy}{dx}} = kx + \frac{k}{2}$, $y(0) = k$,
Now on plotting the diagram we get,



Equation of tangent at $P(x, y)$ is :

$$Y - y = \frac{dy}{dx}(X - x)$$

$$\text{Coordinate of } A = \left(x - y \frac{dx}{dy}, 0\right)$$

$$\text{Coordinate of } B = \left(0, y - x \frac{dy}{dx}\right)$$

$$\therefore (x, y) = \left(\frac{kx - ky \frac{dx}{dy}}{k+1}, \frac{y - x \frac{dy}{dx}}{k+1}\right)$$

$$\text{So, } y = \frac{y - x \frac{dy}{dx}}{k+1}$$

$$\Rightarrow y(k+1) = y - x \frac{dy}{dx}$$

$$\Rightarrow ky = -x \frac{dy}{dx}$$

$$\Rightarrow k \frac{dx}{x} = -\frac{dy}{y}$$

Now integrating both side, we get

$$k \ln |x| + \ln |y| = \ln C \dots\dots(i)$$

Now given equation (i) passes through $(1, 1)$ and $(\frac{1}{10}, 100)$

So, $C = 1$ and $k = 2$

Now putting the value of k , we get

$$e^{\frac{dy}{dx}} = kx + \frac{k}{2}$$

$$\Rightarrow e^{\frac{dy}{dx}} = 2x + 1 \text{ \{as } k = 2\}}$$

$$\Rightarrow \frac{dy}{dx} = \ln(2x + 1)$$

Integrating the above equation we get,

$$y = \frac{1}{2}(2x + 1)(\ln|2x + 1| - 1) + C$$

And the above equation passes through $(0, 2)$

$$\Rightarrow C = \frac{5}{2}$$

$$\text{So, } 2y = (2x + 1)(\ln|2x + 1| - 1) + 5$$

$$\Rightarrow 2y(1) = 3(\ln 3 - 1) + 5$$

$$\Rightarrow 2y(1) = 3 \ln(3) + 2$$

$$\text{Hence, } 4y(1) - 5 \log_e 3 = 2[3 \ln 3 + 2] - 5 \ln 3$$

$$\Rightarrow 4y(1) - 5 \log_e 3 = 6 \ln 3 + 4 - 5 \ln 3$$

$$\Rightarrow 4y(1) - 5 \log_e 3 = 4 + \ln 3 \approx 5$$

Note: This question was bonus in Jee Mains 2023 April session.

8. (2)

Given,

$$\begin{aligned}(1 - x^2 y^2) dx &= y dx + x dy \\ \Rightarrow dx &= \frac{y dx + x dy}{1 - (xy)^2} \\ \Rightarrow dx &= \frac{d(xy)}{1 - (xy)^2} \\ \Rightarrow dx &= \frac{1}{2} \left(\frac{d(xy)}{1 - xy} + \frac{d(xy)}{1 + xy} \right)\end{aligned}$$

Now integrating both side we get,

$$\begin{aligned}\Rightarrow 2x + c &= \ln \left| \frac{1+xy}{1-xy} \right| \\ \Rightarrow \left| \frac{xy+1}{xy-1} \right| &= e^c e^{2x}\end{aligned}$$

Now given, $y(1) = 2$ so putting the value in above equation we get,

$$\begin{aligned}3 &= e^c e^2 \\ \Rightarrow e^c &= \frac{3}{e^2}\end{aligned}$$

Hence, the equation becomes $\left| \frac{xy+1}{xy-1} \right| = 3e^{2x-2}$

Now finding $y(2)$ so putting $x = 2$ in above equation we get,

$$\begin{aligned}\left| \frac{2y+1}{2y-1} \right| &= 3e^2 \\ \Rightarrow 2y+1 &= 2 \cdot 3e^2 y - 3e^2 \\ \Rightarrow 1+3e^2 &= 2y(3e^2 - 1) \\ \Rightarrow y(2) &= \frac{1+3e^2}{2(3e^2-1)}\end{aligned}$$

9. (3)

Given:

$$\frac{dy}{dx} + \frac{5}{x(1+x^5)} y = \frac{(1+x^5)^2}{x^7}$$

This is linear differential equation.

$$\text{I. F.} = e^{\int \frac{5}{x(1+x^5)} dx}$$

$$\Rightarrow \text{I. F.} = e^{\int \frac{5x^5}{x^6(1+x^5)} dx}$$

$$\Rightarrow \text{I. F.} = e^{\int \frac{5}{x^6(x^5+1)} dx}$$

$$\Rightarrow \text{I. F.} = e^{-\int \frac{-5}{x^6(x^5+1)} dx}$$

$$\Rightarrow \text{I. F.} = e^{-\int \frac{d(x^5+1)}{(x^5+1)}}$$

$$\Rightarrow \text{I. F.} = e^{-\log_e(x^5+1)}$$

$$\Rightarrow \text{I. F.} = (x^5 + 1)^{-1} = \frac{x^5}{(x^5+1)}$$

So, solution is given by

$$y \frac{x^5}{(x^5+1)} = \int \frac{x^5}{(x^5+1)} \times \frac{(x^5+1)^2}{x^7} dx$$

$$\Rightarrow \frac{yx^5}{(x^5+1)} = \int \left(\frac{x^5+1}{x^2} \right) dx$$

$$\Rightarrow \frac{yx^5}{(x^5+1)} = \int \left(x^3 + \frac{1}{x^2} \right) dx$$

$$\Rightarrow \frac{yx^5}{(x^5+1)} = \frac{x^4}{4} - \frac{1}{x} + C$$

Since, $y(1) = 2$, so

$$\frac{2}{2} = \frac{1}{4} - 1 + C \Rightarrow C = \frac{7}{4}$$

So,

$$\frac{yx^5}{(x^5+1)} = \frac{x^4}{4} - \frac{1}{x} + \frac{7}{4}$$

Put $x = 2$, then we get

$$\begin{aligned}\frac{y(2) \times 32}{33} &= \frac{16}{4} - \frac{1}{2} + \frac{7}{4} \\ \Rightarrow y(2) &= \frac{21 \times 33}{128} = \frac{693}{128}\end{aligned}$$

10. (1) mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo n

Given,

$$(1+x^2)dy = y(x-y)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{x}{1+x^2}y - \frac{y^2}{1+x^2}$$

$$\Rightarrow \frac{-1}{y^2} \frac{dy}{dx} + \frac{x}{1+x^2} \cdot \frac{1}{y} = \frac{1}{1+x^2}$$

Now let $\frac{1}{y} = t$, we get,

$$\frac{-1}{y^2} y' = \frac{dt}{dx}$$

So, the equation becomes,

$$\frac{dt}{dx} + \frac{x}{1+x^2}t = \frac{1}{1+x^2}$$

$$\text{Now finding, IF} = e^{\int \frac{x}{1+x^2} dx} = \sqrt{1+x^2}$$

So, solution of the differential equation is given by,

$$t\sqrt{1+x^2} = \int \frac{\sqrt{1+x^2}}{1+x^2} dx$$

$$\Rightarrow \frac{1}{y}\sqrt{1+x^2} = \int \frac{1}{\sqrt{1+x^2}} dx$$

$$\Rightarrow \frac{1}{y}\sqrt{1+x^2} = \ln(x + \sqrt{x^2+1}) + C$$

$$\because y(0) = 1 \Rightarrow C = 1$$

$$\Rightarrow \frac{1}{y}\sqrt{1+x^2} = \ln(x + \sqrt{x^2+1}) + 1$$

Now for $x = 2\sqrt{2}$, $y = \beta$ we get,

$$\Rightarrow \frac{3}{\beta} = \ln|2\sqrt{2} + 3| + 1$$

$$\Rightarrow \beta = \frac{3}{1 + \ln|2\sqrt{2} + 3|}$$

$$\Rightarrow 3\beta^{-1} = 1 + \ln|2\sqrt{2} + 3|$$

$$\Rightarrow e^{3\beta^{-1}} = e \cdot e^{\ln(3+2\sqrt{2})}$$

$$\Rightarrow e^{3\beta^{-1}} = e(3 + 2\sqrt{2})$$

11. (1) mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo n

We have been given that $\frac{dy}{dx} = y + 7 \Rightarrow \frac{dy}{y+7} = dx$

$$\Rightarrow \ln|y+7| = x + c$$

$$\Rightarrow |y+7| = k \cdot e^x$$

$$\Rightarrow y = k \cdot e^x - 7$$

$$\Rightarrow y_1(0) = 0 \Rightarrow k = 7 \Rightarrow y_1(x) = 7(e^x - 1)$$

$$\Rightarrow y_2(0) = 1 \Rightarrow 1 = k - 7 \Rightarrow 8 = k$$

$$\Rightarrow y_2(x) = 8e^x - 7$$

$$\Rightarrow y_1(x) = y_2(x) \Rightarrow 8e^x - 7 = 7e^x - 7$$

Hence, No point of intersection.

This is the required option.

12. (6)

Given differential equation is $\frac{dy}{dx} + \frac{4x}{(x^2-1)}y = \frac{x+2}{(x^2-1)^{\frac{5}{2}}}$, $x > 1$

Now IF = $e^{\int \frac{4x}{x^2-1} dx} = (x^2-1)^2$

The required equation will be $\Rightarrow y \cdot (x^2-1)^2 = \int \frac{x+2}{(x^2-1)^{1/2}} dx$

$$\Rightarrow y \cdot (x^2-1)^2 = \frac{1}{2} \int \frac{2x}{(x^2-1)^{1/2}} dx + 2 \int \frac{dx}{(x^2-1)^{1/2}}$$

$$\Rightarrow y \cdot (x^2-1)^2 = 2 \ln(|\sqrt{x^2-1} + x|) + \sqrt{x^2-1} + C$$

Now using, at $x = 2$, $y(2) = \frac{2}{9} \log_e(2 + \sqrt{3})$ we get,

$$\Rightarrow 9 \cdot \frac{2}{9} \ln(2 + \sqrt{3}) = 2 \ln(2 + \sqrt{3}) + \sqrt{3} + C$$

$$\Rightarrow C = -\sqrt{3}$$

Now finding the value of function at $x = \sqrt{2}$ we get,

$$y \times 1 = 2 \ln(1 + \sqrt{2}) + 1 - \sqrt{3}$$

Now on comparing we get,

$$\Rightarrow \beta = 1, \alpha = 2, \gamma = 3$$

$$\Rightarrow \alpha\beta\gamma = 1 \times 2 \times 3 = 6$$

Hence, this is the required answer.

13. (3)

Given,

$$2(y+2)\log_e(y+2)dx + (x+4-2\log_e(y+2))dy = 0$$

Let $x+4 = u$, $y+2 = v$

$dx = du$, $dy = dv$

So, the equation becomes,

$$(2v \ln v)du = -(u-2 \ln v)dv$$

$$\Rightarrow 2v \ln v \frac{du}{dv} + u = 2 \ln v$$

$$\Rightarrow \frac{du}{dv} + \frac{1}{2v \ln v} \cdot u = \frac{1}{v}$$

Which is a linear differential equation,

$$\text{So, IF} = e^{\frac{1}{2} \int \frac{1}{v \ln v} dv} = e^{\frac{1}{2} \ln(\ln v)} = (\ln v)^{\frac{1}{2}}$$

Now solution of differential equation is given by,

$$u \cdot (\ln v)^{\frac{1}{2}} = \int \frac{1}{v} \cdot (\ln v)^{\frac{1}{2}} dv$$

$$\Rightarrow u \cdot (\ln v)^{\frac{1}{2}} = \frac{2}{3} (\ln v)^{\frac{3}{2}} + c \dots \dots (i)$$

Now using given value, $y = e^4 - 2 \Rightarrow x = 1$

$$\therefore v = e^4 \Rightarrow u = 5$$

$$5 \cdot \left(4^{\frac{1}{2}}\right) = \frac{2}{3} \cdot (4)^{\frac{3}{2}} + c$$

$$\Rightarrow 10 = \frac{16}{3} + c$$

$$\Rightarrow c = \frac{14}{3}$$

Now finding, $y = e^9 - 2 \Rightarrow v = y+2 = e^9$

Now putting the value in equation (i) we get,

$$\Rightarrow u \cdot 3 = \frac{2}{3} \times 27 + \frac{14}{3} = 18 + \frac{14}{3}$$

$$\Rightarrow x+4 = u = 6 + \frac{14}{9}$$

$$\Rightarrow x = 2 + \frac{14}{9} = \frac{32}{9}$$